

SystemProfile\Tasks key. Vista also exposes the necessary APIs needed to perform this registrations, thus any third-party multimedia application can sign up to the service. The registry key itself defines areas of multimedia usage that an application might require—and defines task and scheduling preferences in keeping with multimedia type. But how does the whole thing actually work? When a registered application is run, MMCSS—which is run at a high priority itself—takes over and boosts the application's thread into a range indicated by the registry entry, and higher than all other threads running on the system. MMCSS also ensures that such a multimedia application running in the foreground gets 80% of CPU resources. The other 20% are meant to ensure that the entire system does not grind to a stand still while a multimedia application is running. Thus Windows Media Player 11 might assume 8 ms of CPU time over a 10 ms run, while the other 2 ms are doled out to other processes—during this 2 ms of time, the multimedia threads are throttled to lower priorities by the MMCSS service.

Thread Priorities

Thread priorities in Windows range from 0 to 31: thread priorities 16 and higher are in the real-time priority range and higher than all other threads on a system. Only the kernel's memory manager threads work at higher priorities of 28 and 29. MMCSS runs its priority-management thread at priority 27.

3.3 File-based Symbolic Links

A feature long present under UNIX operating systems finally makes its presence under Vista: symbolic links. A symbolic link is essentially a pointer to the actual location of a data. Under Windows 2000, directories had symbolic links and were called directory junctions, but it is with the advent of Vista that this concept has been extended to the files themselves. When a user removes the symbol-